

Name: _____

Class: _____



Mathematics Extension 2

Task 2, Term 1, 2020

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 6 –10, show relevant mathematical reasoning and/or calculations

Total marks: 80

Section I – 5 marks (pages 2-3)

- Attempt Questions 1–5
- Allow about 7 minutes for this section

Section II – 75 marks (pages 4-8)

- Attempt Questions 6-10
- Allow about 1 hours and 53 minutes for this section

Section I (5 marks)

Write your answers on the multiple-choice answer sheet provided.

1. Given that $(1 - 2i)^2 = -3 - 4i$, the roots of the equation: $z^2 - 5z + (7 + i) = 0$ are:

A. $3 - i, 2 - i$

C. $3 - i, 2 + i$

B. $3 + i, 2 - i$

D. $3 + i, 2 + i$

2. $x = 3 + 4i, y = -1 + 2i$, find the value of $|x - \bar{y}|$.

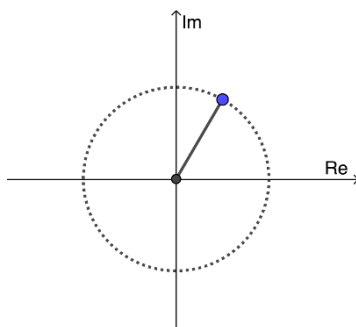
A. $15 + \sqrt{5} + 20i$

C. $10 + 20i$

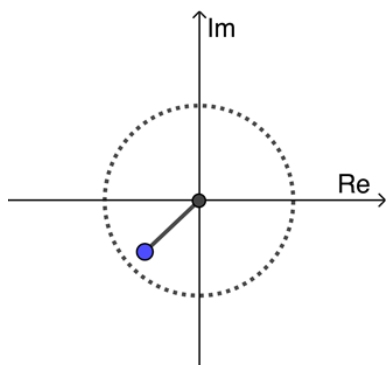
B. $20 + 20i$

D. $15 - \sqrt{5} + 20i$

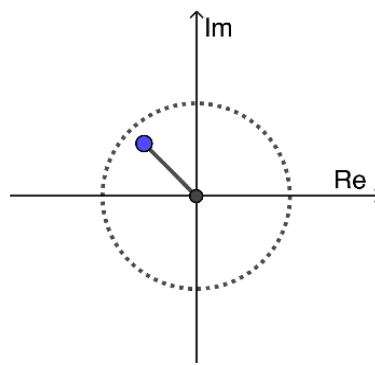
3. The Argand Diagram below shows the complex number $z = x + iy$. Which diagram best represents the complex number $iz + \bar{z}$?



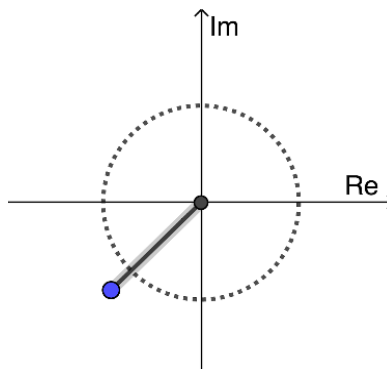
A.



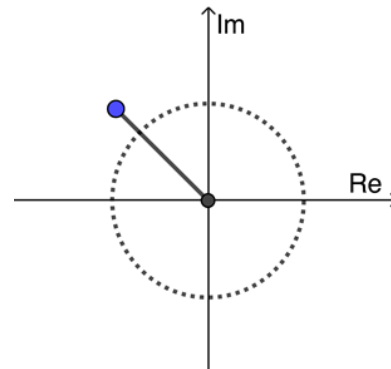
B.



C.



D.



- Section II begins on the next page.**

Section II (75 marks)

Attempt Questions 6-10

Allow about 1 hour and 53 minutes for this section

Start each question on a new page. Extra paper is available.

In Questions 6-10, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (15 marks)

Start a new page.

- a) Find all the zeros of $P(x) = x^4 - x^3 - 2x^2 + 6x - 4$ over $\mathbf{C}$, given that $(1 - i)$ is one of the zeros. 3
- b) (i) Express $3 + \sqrt{3}i$ and $3 - \sqrt{3}i$ in the Modulus Argument Form. 2
- (ii) Find the values of n that satisfy $(\frac{1}{3 + \sqrt{3}i})^n + (\frac{1}{3 - \sqrt{3}i})^n = 0$. 3
- c) If ω is a non-real cube root of unity, show that $(1 + \omega)^3(1 + 2\omega + 2\omega^2) = 1$. 4
- d) (i) Sketch, without calculating stationary points, the graph of the function $f(x)$ given by $f(x) = (x - p)(x - q)(x - r)$, where $p < q < r$ and $p, q, r \in \mathbb{R}$. 1
- (ii) By considering the equation $f'(x) = 0$, show that $(p + q + r)^2 > 3(qr + rp + pq)$. 2

Question 7 begins on the next page.

Question 7 (15 marks)

Start a new page.

a) Sketch the following loci on separate Argand diagrams.

(i) $\text{Arg}(z + i) = \text{Arg}(z - 1)$ 2

(ii) $z - \bar{z} \leq |z|^2$ 3

b) If $z_1 = 3e^{\frac{i\pi}{6}}$, $z_2 = 2e^{\frac{i\pi}{6}}$, find: 2

(i) Find $\bar{z}_2$ in the exponential form.

(ii) Find $\frac{z_2}{z_1} - z_1\bar{z}_2 + \frac{4}{z_2}$ in the form of $x + iy$. 3

c) Find the area of a triangle whose vertices are z , iz , and $z + iz$. 2

d) Let a and b be rationals and x irrational. Show that if $\frac{x+a}{x+b}$ is rational then $a = b$. 3

Question 8 begins on the next page.

Question 8 (15 marks)

Start a new page.

- a) Given that $\frac{1+7i}{1+2i} = re^{i\theta}$, $r > 0$ and $-\pi < \theta \leq \pi$, find the value of r and θ (Provide the value of θ rounded to two decimal places). 2

- b) $OPQR$ is a parallelogram on an Argand Diagram. The point O is the origin, and the point P and Q represent the complex number z_1 and z_2 respectively.

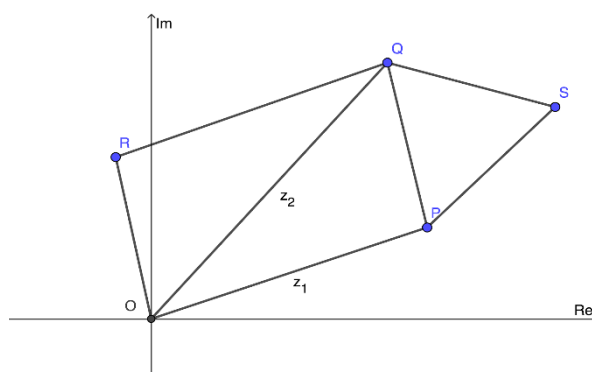
Given $|z_2| = \frac{2\sqrt{3}}{3}|z_1|$, $\arg\left(\frac{z_2}{z_1}\right) = \frac{\pi}{6}$, $0 < \arg(z_1) < \arg(z_2) < \frac{\pi}{2}$,

- (i) Prove $z_2 = \left(1 + \frac{\sqrt{3}}{3}i\right)z_1$. 2

- (ii) Prove $|z_2 - z_1| = \frac{1}{2}|z_2|$. 1

- (iii) Find $\arg\left(\frac{z_2 - z_1}{z_1}\right)$. 3

- (iv) Given that ΔQPS is an equilateral triangle and the point S represents complex number z_3 . Find the expression of z_3 in terms of z_1 . 3



- (c) (i) Prove the following for real and positive numbers a, b . 1

$$\frac{a+b}{2} \geq \sqrt{ab}$$

- (ii) Hence, show that $\sqrt{a} \sqrt{b+c} \leq \frac{a+b+c}{2}$. 1

- (iii) Show that $\sqrt{\frac{a}{b+c}} + \sqrt{\frac{b}{c+a}} + \sqrt{\frac{c}{a+b}} > 2$ for $a, b, c > 0$. 2

Question 9 begins on the next page.

Question 9 (15 marks)

Start a new page.

- a) (i) Express $\cos 5\theta$ in the form of $a\cos^5\theta + b\cos^3\theta + c\cos\theta$, giving the numerical values of a, b, c . 3
- (ii) Find all the solutions of the equation $z^5 + 1 = 0$. 2
- (iii) Sketch all the solutions of the equation $z^5 + 1 = 0$ on a unit circle. 1
- (iv) Express $z^5 + 1$ as a product of linear and quadratic factors with real coefficients. 2
- (v) Find the value of $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} + \cos \frac{7\pi}{5} + \cos \frac{9\pi}{5}$, giving reasons. 2
- b) Let $x, y \in \mathbb{Z}$. Prove that if $x^2(y^2 - 2y)$ is odd, then x and y are both odd. 2
- c) Prove that a number is divisible by 8 if and only if the last three digits of a number are divisible by 8. 3

Question 10 begins on the next page.

Question 10 (15 marks)

Start a new page.

- a) (i) Factorise $z^4 + 1 = 0$ into quadratic factors of real coefficients. **1**
- (ii) Divide by z^2 and hence show $\cos 2\theta = 2 \left(\cos \theta - \cos \frac{\pi}{4} \right) \left(\cos \theta - \cos \frac{3}{4}\pi \right)$. **4**
- b) If the roots of the equation $z^4 + az^3 + (12 + 9i)z^2 + bz = 0$ are the vertices of a square, and $2b - 13a = x$, find the values of x , where $x \in \mathbb{Z}$ and a and b are complex numbers. **3**
- c) Given a, b, c are positive real numbers with $a > b$ and $c^2 > ab$, prove by contradiction that $\frac{a+c}{\sqrt{a^2+c^2}} - \frac{b+c}{\sqrt{b^2+c^2}} > 0$. **4**
- d) Given $\frac{x+y+z}{3} \geq \sqrt[3]{xyz}$, show that $\frac{1}{A} + \frac{1}{B} + \frac{1}{C} \geq \frac{9}{\pi}$ if A, B, C are three interior angles (in radians), of $\triangle ABC$. **3**

END OF EXAMINATION

MATHEMATICS Extension 2: Question.....6			1/4
Suggested Solutions	Marks Awarded	Marker's Comments	
a) Find all zeros $P(x) = x^4 - x^3 - 2x^2 + 6x - 4 \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{given}$ $(1-i)$ is zero $\therefore$ the conjugate root is $(1+i)$ — ① $\therefore (1-i)(1+i) = x^2 - 2x + 2$ $\begin{array}{r} x^2 + x - 2 \\ x^2 - 2x + 2 \overline{) x^4 - x^3 - 2x^2 + 6x - 4} \\ \underline{-x^4 - 2x^3 + 2x^2} \downarrow \\ x^3 - 4x^2 + 6x \downarrow \\ \underline{-x^3 - 2x^2 + 2x} \downarrow \\ -2x^2 + 4x - 4 \downarrow \\ \underline{-(-2x^2 + 4x - 4)} \\ 0 \end{array}$ $P(x) = (x+2)(x-1)(x-(1+i))(x-(1-i))$ } ① zeros = $(1-i), (1+i), -2, 1$		zeros $\neq$ factored form But! No mark was penalised.	
b) i) $3 + \sqrt{3}i$ and $3 - \sqrt{3}i$ in mod arg form. mod = $3^2 + (\sqrt{3})^2$ $= \sqrt{12} = 2\sqrt{3}$ arg = $\tan^{-1} \frac{\sqrt{3}}{3} = \frac{\pi}{6}$ $2\sqrt{3} \cos(\frac{\pi}{6}) + i \sin(\frac{\pi}{6})$ — ① $2\sqrt{3} \cos(-\frac{\pi}{6}) + i \sin(-\frac{\pi}{6})$ } — ① or $2\sqrt{3} \cos(\frac{\pi}{6}) - i \sin(\frac{\pi}{6})$		Unfortunately only 2 marks allocated. So the marking was based on correct arg for both and correct mod for both. If not marked the way student would have lost both marks. - pls define cis	

Suggested Solutions

Marks Awarded

Marker's Comments

b) ii) $\left(\frac{1}{3+\sqrt{3}i}\right)^n + \left(\frac{1}{3-\sqrt{3}i}\right)^n = 0$

$$= \left(2\sqrt{3} \operatorname{cis} \frac{\pi}{6}\right)^{-n} + \left[2\sqrt{3} \operatorname{cis} \left(-\frac{\pi}{6}\right)\right]^{-n} \quad \text{by DM7 (1)}$$

$$= (2\sqrt{3})^{-n} \left(\operatorname{cis} -\frac{n\pi}{6} + \operatorname{cis} \frac{n\pi}{6}\right)$$

$$= (2\sqrt{3})^{-n} \left(\cos \frac{n\pi}{6} - i \sin \frac{n\pi}{6} + \cos \frac{n\pi}{6} + i \sin \frac{n\pi}{6}\right)$$

$$= (2\sqrt{3})^{-n} \times 2 \cos\left(\frac{n\pi}{6}\right) = 0 \quad \text{(1)}$$

$$\cos\left(\frac{n\pi}{6}\right) = \cos(0)$$

$$\frac{n\pi}{6} = \pm \frac{\pi}{2} + 2k\pi \quad \text{(1)}$$

$$n = \pm 3$$

$n = 6k+3, k \in \mathbb{Z}$
(or $12k \pm 3$)

c) ω is non-real cube root of unity } given & show
show $(1+\omega)^3(1+2\omega+2\omega^2) = 1$

METHOD 1

$\omega^3 = 1$

but.

$(\omega-1)(1+\omega+\omega^2) = 0 \quad \omega \neq 1 \quad \text{(1)}$
 $1+\omega = -\omega^2$
 $\omega+\omega^2 = -1$

LHS:

$$(1+\omega)^3(1+2\omega+2\omega^2)$$

$$(1+\omega)^3(1+\omega+\omega+\omega^2+\omega^2)$$

$$(-\omega^2)^3(1+\omega+\omega^2+\omega+\omega^2)$$

$= (-1)(0-1) \quad \text{(1)}$

Lucky!!

- didn't penalise for not mentioning cis for DM7

note*

have to consider even and odd func. at this step.

$\sin(-\theta) = -\sin \theta$

$\cos(-\theta) = \cos \theta$

one could have written as +ve or -ve of the function and was marked accordingly. marks was not aware if ± was not mention but cfe was awarded in a question.

Too many marks allocated for this question.

(Could have given a mark for showing that ω^2 is the other non real root. it is otherwise assumed but why should it be?)

(But you only used it in method 2!) Or is it method 3!)

$= 1$
 $= \text{RHS}$

Suggested Solutions

Marks Awarded

Marker's Comments

METHOD 2

$$\text{LHS } (1+\omega)^3(1+2\omega+2\omega^2) = 1$$

$$(1+3\omega+3\omega^2+\omega^3)(1+2\omega+2\omega^2) \text{ --- } \textcircled{1}$$

$$= 1+3\omega+3\omega^2+\omega^3+2\omega+6\omega^2+6\omega^3+2\omega^4$$

$$+ 2\omega^2+6\omega^3+6\omega^4+2\omega^5$$

$$= 1+3\omega+2\omega+3\omega^2+6\omega^2+2\omega^2+\omega^3+6\omega^3$$

$$+ 6\omega^3+2\omega^4+6\omega^4+2\omega^5$$

$$= 1+5\omega+11\omega^2+13\omega^3+8\omega^4+2\omega^5$$

$$= 1+5\omega+11\omega^2+13+2\omega+2\omega^2 \rightarrow \text{cos } \omega^3=1$$

$$= 14+5\omega+8\omega+11\omega^2+2\omega^2 \text{ --- } \textcircled{1}$$

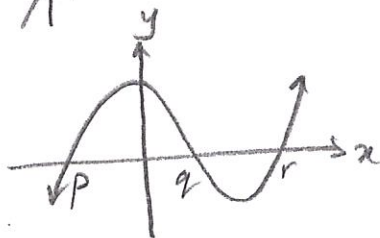
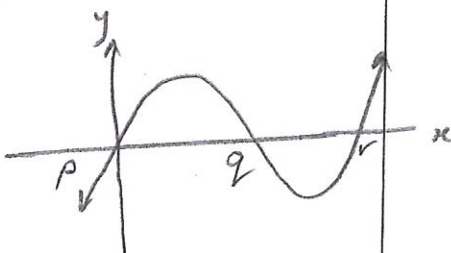
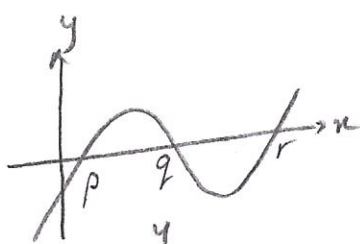
$$= 14+13(\omega+\omega^2)$$

$$= 14+13(-1)$$

$$= 1$$

$$= \text{RHS}$$

d) i) sketch $f(x)=(x-p)(x-q)(x-r)$
 $p < q < r, p, q, r \in \mathbb{R}$



$$14\omega+\omega^2=0$$

$$\therefore \omega+\omega^2=-1$$

} --- $\textcircled{1}$

$\textcircled{1}$

any was accepted
 with roots in
 correct order
 and +ve cubic
 function.

MATHEMATICS: Question.....

4/4

Suggested Solutions

Marks Awarded

Marker's Comments

dii) by considering $f'(x) = 0$
show $(p+q+r)^2 > 3(qr+rp+pq)$

$$f(x) = (x-p)(x-q)(x-r) \quad \text{--- expand out}$$

$$= x^3 - x^2(p+q+r) + x(pq+qr+pr) - pqr$$

$$f'(x) = 3x^2 - 2x(p+q+r) + (pq+qr+pr) \quad \text{--- (1)}$$

for stationary pts $f'(x) = 0$
there are 2 distinct stnary pts.

$$\text{ie } \Delta > 0$$

$$\Delta = b^2 - 4ac$$

$$= 4(p+q+r)^2 - 4(3)(pq+qr+pr) > 0 \quad \text{--- (1)}$$

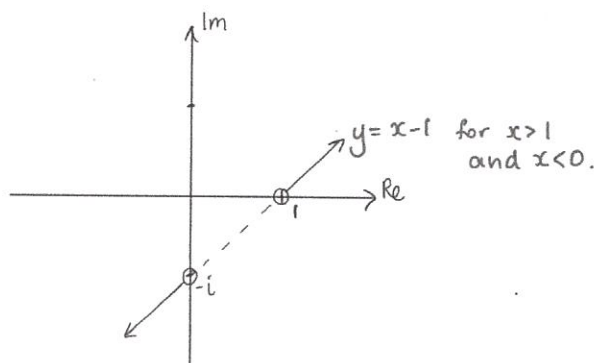
$$4(p+q+r)^2 > 12(pq+qr+pr)$$

$$(p+q+r)^2 > 3(pq+qr+pr) \quad \text{shown}$$

< Question 7 solutions

Sunday, 19 April 2020 8:39 pm

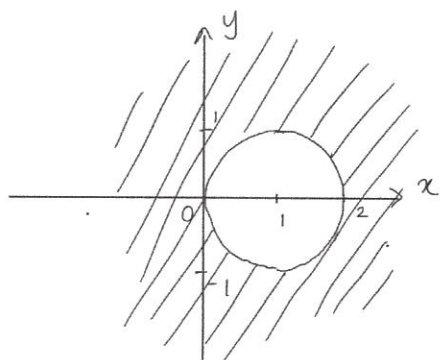
(i)



① intercepts and arrows in correct direction

① showing line is $y = x - 1$
by - labelling the line
or - showing dotted line between intercepts
- showing angles between line and axes.

$$\begin{aligned} \text{(ii)} \quad z + \bar{z} &\leq |z|^2 \\ x + iy + (x - iy) &\leq (x^2 + y^2) \\ 2x &\leq (x^2 + y^2) \\ x^2 + y^2 - 2x &\geq 0 \\ (x - 1)^2 + y^2 &\geq 1 \end{aligned}$$



[Some students were not aware of the correction from $z - \bar{z}$ to $z + \bar{z}$
- no penalty for no diagram when $z - \bar{z}$ used]

① Inequality with $2x$
① $x^2 + y^2$
① correct diagram

$$\text{b)} \quad z_1 = 3e^{i\pi/6}, \quad z_2 = 2e^{i\pi/6}$$

$$\begin{aligned} \text{(i)} \quad z_2 &= 2e^{i\pi/6} \\ &= 2(\cos \pi/6 + i \sin \pi/6) \\ \bar{z}_2 &= 2(\cos \pi/6 - i \sin \pi/6) \\ &= 2(\cos(-\pi/6) + i \sin(-\pi/6)) \\ &= 2e^{-i\pi/6} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \frac{z_2}{z_1} - z_1 \bar{z}_2 + \frac{4}{z_2} \\ &= \frac{2e^{i\pi/6}}{3e^{i\pi/6}} - 3e^{i\pi/6} \times 2e^{-i\pi/6} + \frac{4}{2e^{i\pi/6}} \\ &= \frac{2}{3} - 6 + 2e^{-i\pi/6} \\ &= -5\frac{1}{3} + 2(\cos(-\pi/6) + i \sin(-\pi/6)) \\ &= -5\frac{1}{3} + 2(\cos \pi/6 - i \sin \pi/6) \\ &= (-5\frac{1}{3} + \sqrt{3}) - i \\ &= \frac{3\sqrt{3} - 16}{3} - i \end{aligned}$$

Bald answer ②

①
①

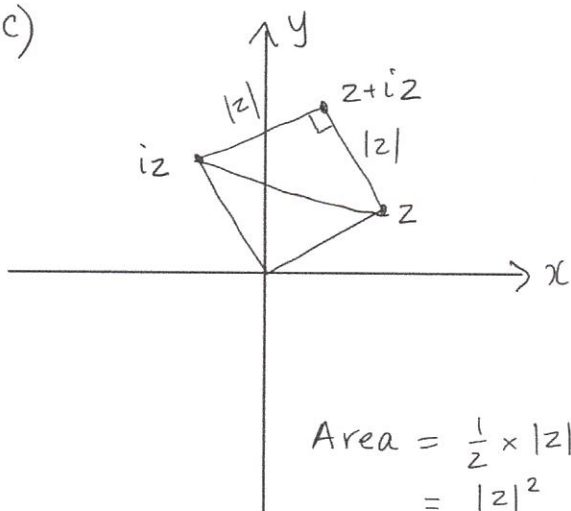
① for $\frac{z_2}{z_1} = \frac{2}{3}$

① for $z_1 \bar{z}_2 = 6$

① for $\frac{4}{z_2} = \sqrt{3} - i$

<

c)



$$\begin{aligned} \text{Area} &= \frac{1}{2} \times |z| \times |z| \\ &= \frac{|z|^2}{2} \text{ square units} \end{aligned}$$

① lengths of sides

① area expression

d) Method 1: ContradictionSuppose $\frac{x+a}{x+b}$ is rationalie. $\frac{x+a}{x+b} = r$ where r is rationalAssume $r \neq 1$

$$\frac{x+a}{x+b} = r$$

$$x+a = rx+rb$$

$$x(1-r) = rb-a$$

$$x = \frac{rb-a}{1-r}$$

But a, b, r are rationals $\therefore x$ is rationalBut x is irrational $\therefore r=1$ by contradiction, which is rational $\therefore \frac{x+a}{x+b} = 1$ which is rational

$$x+a = x+b$$

 $\therefore a=b$ if $\frac{x+a}{x+b}$ is rational.

Method 2: Contradiction

If $\frac{x+a}{x+b}$ is rational and a, b are rational

then $\frac{x+a}{x+b} = \frac{p}{q}$ where $p, q \in \mathbb{Z}$. Assume $p \neq q$

$$qx + qa = px + pb$$

$$qx - px = pb - qa$$

$$x(q-p) = pb - qa$$

but x is irrational

$$x = \frac{pb - qa}{q-p}$$

if $a \neq b$ then $\frac{q-p}{q-p} x$ is rational as $\frac{pb - qa}{q-p}$ is rational

if $a = b$ then $x = \frac{a(p-q)}{q-p}$
 $= -a$ which is rational

but x is irrational

$\therefore p = q$ by contradiction of assumption

Now if $p = q$ then $\frac{p}{q} = 1$

$$\frac{x+a}{x+b} = 1$$

$$\text{So } x+a = x+b$$

$$\therefore a = b$$

$$\text{or } x = \frac{b(p-q)}{q-p}$$
$$= -b$$

$$\therefore -a = -b$$
$$a = b$$

① expression for x
following assumption $p \neq q$

① show by contradiction that $p = q$

① show $a = b$

Note: only 1 mark if $p = q$ not considered.

Method 3: Using Contrapositive and Contradiction

Prove by contrapositive if $a \neq b$ then $\frac{x+a}{x+b}$ is irrational

$a \neq b$, a, b are rational

$$\frac{x+a}{x+b} = \frac{x+b}{x+b} - \frac{b-a}{x+b}$$
$$= 1 - \frac{b-a}{x+b}$$

① stating contrapositive
and finding x

Consider $\frac{b-a}{x+b}$. For this to be rational $b-a = 0$,

Since $\frac{x+a}{x+b}$ is irrational.

But $a \neq b$ in assumption

① contradiction

$\therefore \frac{b-a}{x+b}$ is irrational

$\therefore \frac{x+a}{x+b}$ is irrational if $a \neq b$ ① proof of contrapositive

$\therefore$ If $\frac{x+a}{x+b}$ is rational then $a = b$ by contrapositive

Method 4: Equating rational & irrational parts

If $\frac{x+a}{x+b}$ is rational let $\frac{x+a}{x+b} = \frac{p}{q}$ $p, q \in \mathbb{Z}$

$$q(x+a) = p(x+b)$$

$$qx + qa = px + pb \quad (1)$$

Since a, b, p, q are rational qa and pb are rational

Equating irrational parts of same form

$$qx = px$$

$$\therefore p = q$$

$$\therefore \frac{x+a}{x+b} = 1$$

$$x+a = x+b$$

$$\therefore a = b$$

① Recognising irrational parts

① Showing $p = q$

① Showing $a = b$

Suggested Solutions

Marks

Marker's Comments

a) Method 1

$$\frac{1+7i}{1+2i} = \frac{(1+7i)(1-2i)}{(1+2i)(1-2i)}$$

$$= \frac{1+14+5i}{5}$$

$$= \frac{15+5i}{5}$$

$$= 3+i$$

$$\therefore r = \sqrt{3^2+1^2} = \underline{\underline{\sqrt{10}}}$$

$$\theta = \tan^{-1}\left(\frac{1}{3}\right) = \underline{\underline{0.32}} \text{ (to 2 DP)}$$

Method 2

$$\frac{1+7i}{1+2i} = \frac{\sqrt{50} e^{i(\tan^{-1}7)}}{\sqrt{5} e^{i(\tan^{-1}2)}}$$

$$= \sqrt{10} e^{i(\tan^{-1}7 - \tan^{-1}2)}$$

$$= \sqrt{10} e^{i(0.32)}$$

$$\therefore \underline{\underline{r = \sqrt{10}}} \quad \underline{\underline{\theta = 0.32}}$$

Too many people left argument in degrees where radians are clearly required in question.

Usually 1 mark for $\sqrt{10}$ and 1 for 0.32. Where a slip was made in the initial calculation, 1 mark if rest of answer followed.

b) i)

$$\arg\left(\frac{z_2}{z_1}\right) = \frac{\pi}{6} \quad 0 \leq \arg z_1 < \arg z_2 < \frac{\pi}{2}$$

$$\therefore \arg z_2 = \arg z_1 + \frac{\pi}{6}$$

$$z_2 = |z_2| e^{i(\arg z_1 + \pi/6)}$$

$$= \frac{2\sqrt{3}}{3} |z_1| e^{i \arg z_1} \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

$$= z_1 \cdot \frac{2\sqrt{3}}{3} \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

$$= z_1 \left(\frac{2\sqrt{3}}{3} \left(\frac{\sqrt{3}}{2} + \frac{i}{2} \right) \right) \quad *$$

$$= z_1 \left(1 + \frac{\sqrt{3}}{3} i \right)$$

1 for modulus and argument shifts.
1 for final answer.

Several people lost second mark by omitting last line of working*. Essential when result is given.

Suggested Solutions

Marks

Marker's Comments

$$b) \text{ ii) } z_2 - z_1 = \frac{\sqrt{3}i}{3} z_1 \text{ (Using (i))}$$

$$\therefore |z_2 - z_1| = \left| \frac{\sqrt{3}}{3} \right| |z_1|$$

$$= \frac{\sqrt{3}}{3} \cdot \frac{3}{2\sqrt{3}} |z_2| \text{ (Since } |z_1| = \frac{2\sqrt{3}}{3} |z_2| \text{)}$$

$$= \frac{|z_2|}{2}$$

1

Again several people did not satisfactorily explain the last line. Essential when result given

$$\text{iii) } \arg\left(\frac{z_2 - z_1}{z_1}\right) = \arg(z_2 - z_1) - \arg(z_1)$$

$$= \arg\left(z_1 + \frac{\sqrt{3}i}{3} z_1 - z_1\right) - \arg(z_1) \text{ (Using (i))}$$

$$= \arg\left(\frac{\sqrt{3}i}{3}\right) + \arg(z_1) - \arg(z_1)$$

$$= \arg\left(\frac{\sqrt{3}i}{3}\right)$$

$$= \pi/2 \text{ as this is purely imag.}$$

2

Not really worth 3 marks but take what you can!

A lot of students used a longer geometrical approach ending with an answer of $\pi/2$ but with

iv)

$$\vec{OS} = \vec{OP} + \vec{PS}$$

$$z_3 = z_1 + (z_2 - z_1) e^{-i\pi/3}$$

$$= z_1 + \frac{\sqrt{3}i}{3} \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right) z_1$$

$$= z_1 + \frac{\sqrt{3}i}{3} \left(\frac{1}{2} - i \frac{\sqrt{3}}{2} \right) z_1$$

$$= \left(1 + \frac{\sqrt{3}i}{6} + \frac{3}{6} \right) z_1$$

$$= \left(\frac{3}{2} + \frac{\sqrt{3}i}{6} \right) z_1$$

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no reasoning why the argument was not $-\pi/2$. They generally got 2 marks of the 3.

1 for the (implied) vector addition

1 for a correct rotation in the equilateral triangle

1 for getting to the final answer.

*See note on next page

Suggested Solutions

Marks

Marker's Comments

* There was an awful mixture of notations in (b) and especially part (iv).

Vectors and complex numbers should not be mixed

$$\text{e.g. } \vec{OS} = z_1 + \vec{PS} \quad (\text{NO})$$

Others used S when they meant $\vec{OS}$. And all of these things were mixed up at random. Generally I marked what they would have intended but it went a bit against the grain.

$$\text{c) i) } (\sqrt{a} - \sqrt{b})^2 \geq 0$$

$$a + b - 2\sqrt{a}\sqrt{b} \geq 0$$

$$\therefore (a+b)/2 \geq \sqrt{ab} \quad (\text{since } a > 0, b > 0)$$

$$\text{ii) So } (x+y)/2 \geq \sqrt{x}\sqrt{y} \quad (x, y > 0)$$

$$\text{Substitute } x=a \quad y=b+c \quad (a, b, c > 0)$$

$$\therefore (a+b+c)/2 \geq \sqrt{a}\sqrt{b+c}$$

$$\text{or } \sqrt{a}\sqrt{b+c} \leq \frac{a+b+c}{2}$$

$$\text{iii) } \sqrt{a}\sqrt{b+c} \leq \frac{a+b+c}{2}$$

Take reciprocals,

$$\frac{1}{\sqrt{a}\sqrt{b+c}} \geq \frac{2}{a+b+c} \quad (\text{all } > 0)$$

$$\frac{\sqrt{a}}{\sqrt{b+c}} \geq \frac{2a}{a+b+c} \quad (1)$$

$$\text{Similarly } \frac{\sqrt{b}}{\sqrt{c+a}} \geq \frac{2b}{a+b+c} \quad (2)$$

$$\frac{\sqrt{c}}{\sqrt{a+b}} \geq \frac{2c}{a+b+c} \quad (3)$$

Only a handful of students took the reciprocal. Most got bogged down in algebra. The first mark was for taking a useful reciprocal.

c) iii) (Continued)

Add lines ①, ② & ③

$$\frac{\sqrt{a}}{\sqrt{b+c}} + \frac{\sqrt{b}}{\sqrt{c+a}} + \frac{\sqrt{c}}{\sqrt{a+b}} \geq \frac{2a+2b+2c}{(a+b+c)}$$

$$\frac{\sqrt{a}}{\sqrt{b+c}} + \frac{\sqrt{b}}{\sqrt{c+a}} + \frac{\sqrt{c}}{\sqrt{a+b}} \geq 2$$

But equality holds in original statement only when $a=b+c$.
The other parts require $b=c+a$
 $c=a+b$

These are only simultaneously possible when $a=0, b=0, c=0$ which is not the case since $a, b, c > 0$. Therefore the final sign is strictly $>$.

$$\frac{\sqrt{a}}{\sqrt{b+c}} + \frac{\sqrt{b}}{\sqrt{c+a}} + \frac{\sqrt{c}}{\sqrt{a+b}} > 2$$

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Getting this far got the second mark. (Could have done with a 3rd mark)

Two people managed to explain the $>$, others didn't try. Unfortunately there was no mark left.

Q9a

JRAMS T1 Ex+2

1/4

a) Let $c = \cos \theta$ $s = \sin \theta$

both lines $\left\{ \begin{aligned} (\cos \theta)^5 &= c^5 + 5c^4 is + 10c^3(is)^2 + 10c^2(2s)^3 + 5c(is)^4 + (is)^5 \\ &= (c^5 - 10c^3s^2 + 5cs^4) + i(5c^4s - 10c^2s^3 + s^5) \end{aligned} \right.$

$$(cis \theta)^5 = \cos 5\theta + i \sin 5\theta \quad (\text{De Moivre's Th})$$

Equating real coefficient

$$\begin{aligned} \cos 5\theta &= c^5 - 10c^3s^2 + 5cs^4 \\ &= c^5 - 10c^3(1-c^2) + 5c(1-c^2)^2 \\ &= c^5 - 10c^3 + 10c^5 + 5c(1 - 2c^2 + c^4) \\ &= c^5 - 10c^3 + 10c^5 + 5c - 10c^3 + 5c^5 \\ &= 16c^5 - 20c^3 + 5c \end{aligned}$$

Answer $= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$

ii) $z^5 = -1 = cis(\pi + 2n\pi) \quad n \in \mathbb{Z}$

Let $z = cis \theta$

$$z^5 = cis 5\theta \quad (\text{De Moivre's Th})$$

$$\therefore 5\theta = \pi + 2n\pi$$

$$\theta = \frac{\pi + 2n\pi}{5}$$

$$\therefore z = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}, \quad \cos^{-\frac{\pi}{5}} + i \sin(-\frac{\pi}{5})$$

Answer $\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}, \quad \cos^{-\frac{3\pi}{5}} + i \sin(-\frac{3\pi}{5})$
 $\cos \pi + i \sin \pi \quad (= -1)$

9b) Method 1 : using contrapositive

∴ If either x or y is even, then $x^2(y^2 - 2y)$ is even

If x is even, let $x = 2m$, $m \in \mathbb{Z}$

$$x^2(y^2 - 2y) = (2m)^2(y^2 - 2y)$$

$$x^2(y^2 - 2y) = 4(2m^2)(y^2 - 2y)$$

$$x^2(y^2 - 2y) = 2k \quad k \in \mathbb{Z}$$

∴ even

If y is even, let $y = 2n$, $n \in \mathbb{Z}$

$$x^2(y^2 - 2y) = x^2[4n^2 - 4n]$$

$$= 2 \times 2 [x^2] [n^2 - n]$$

∴ $x^2(y^2 - 2y)$ is even

Alternatively students can prove

- | | | |
|--------|------------------|----------|
| Case 1 | x odd | y even |
| 2 | x even | y odd |
| 3 | x, y both even | |

if only 2 of
these 3 cases
only 1 m

Method 2 : Direct Proof

∴ If $x^2(y^2 - 2y)$ is odd, then x and y are both odd

A product of 2 number is only odd
only if both are odd

∴ x^2 is odd, $y^2 - 2y$ is odd

for x odd

But $x^2 \equiv x \times x$, $y^2 - 2y = y(y - 2)$

∴ both x, y need to be odd

If y is odd $y - 2$ is also odd

for y odd

∴ x, y are both odd

★ No marks given to converse

9c) To prove $P \iff Q$, need to prove $P \Rightarrow Q$, and
prove $Q \Rightarrow P$

To prove $P \Rightarrow Q$:

Let the number $N = 1000a + 100b + 10c + d$, $a, b, c, d \in \mathbb{Z}$
 $= 8(125a) + 100b + 10c + d$ ★

If N is divisible by 8, say $N = 8m$, $m \in \mathbb{Z}$

$$8m = 8(125a) + 100b + 10c + d$$

$$8(m - 125a) = 100b + 10c + d$$

Since $m - 125a$ is an integer

$\therefore 100b + 10c + d$ is divisible by 8

To prove $Q \Rightarrow P$:

If the last 3 digits is divisible by 8,

$$\text{Let } 100b + 10c + d = 8p \quad p \in \mathbb{Z}$$

$$\text{From } \star \quad N = 8(125a) + 8p$$

$$N = 8(125a + p)$$

$$N = 8q \quad q \in \mathbb{Z}$$

$\therefore N$ will be divisible by 8

Note (1) must see $1000a = 8 \times 125a$ to get any mark

(2) Some students have misunderstood
each of the last 3 digits has to be
divisible by 8. Om.

(3) some forgot to define new variables,
if $m \in \mathbb{Z}$

MATHEMATICS Extension 2: Question 10...

Suggested Solutions

Marks

Marker's Comments

a) i) $z^4 + 1 = 0$

$$z^4 + 2z^2 + 1 - 2z^2 = 0$$

$$(z^2 + 1)^2 - 2z^2 = 0$$

$$(z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1) = 0$$

ii) $z^4 + 1 = (z^2 - 2\cos\frac{\pi}{4}z + 1)$

$$(z^2 - 2\cos\frac{3\pi}{4}z + 1)$$

$$z^2 + \frac{1}{z^2} = (z - 2\cos\frac{\pi}{4} + \frac{1}{z})$$

$$(z - 2\cos\frac{3\pi}{4} + \frac{1}{z})$$

let $z = \cos\theta + i\sin\theta$

$$z^n = \cos n\theta + i\sin n\theta \text{ (De Moivre's Theorem)}$$

$$z^{-n} = \cos(-n\theta) + i\sin(-n\theta) \text{ (De Moivre's Theorem)}$$

$$= \cos n\theta - i\sin n\theta$$

$$\therefore z^n + z^{-n} = \cos n\theta + i\sin n\theta + \cos n\theta - i\sin n\theta$$

$$= 2\cos n\theta$$

$$\therefore z^2 + z^{-2} = 2\cos 2\theta$$

$$z + z^{-1} = 2\cos\theta$$

$$\therefore 2\cos 2\theta = (2\cos\theta - 2\cos\frac{\pi}{4})$$

$$(2\cos\theta - 2\cos\frac{3\pi}{4})$$

$$2\cos 2\theta = 4(\cos\theta - \cos\frac{\pi}{4})$$

$$(\cos\theta - \cos\frac{3\pi}{4})$$

$$\therefore \cos 2\theta = 2(\cos\theta - \cos\frac{\pi}{4})$$

$$(\cos\theta - \cos\frac{3\pi}{4})$$

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changing $\sqrt{2}$
to correct
values

dividing
correctly

showing these
are true.

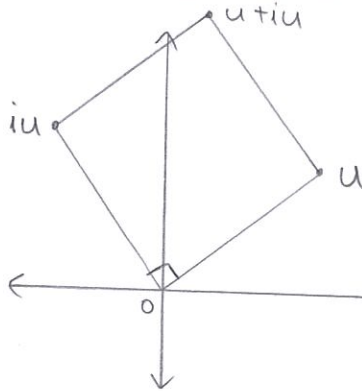
correct
calculation.

$$b) z^4 + az^3 + (12+9i)z^2 + bz = 0$$

$$z(z^3 + az^2 + (12+9i)z + b) = 0$$

$\therefore 0$ is a root.

$\therefore 0$ is a vertex of the square.



$\therefore$ the other vertices are:

u , iu and $u+iu$

for the equation

$$z^3 + az^2 + (12+9i)z + b = 0$$

sum of roots two at a time:

$$u(iu) + u(u+iu) + iu(u+iu) = 12+9i$$

$$iu^2 + u^2 + iu^2 - u^2 = 12+9i$$

$$3iu^2 = 12+9i$$

$$u^2 = 3-4i$$

$$\therefore u = \pm(2-i)$$

sum of roots:

$$u+iu+u+iu = -a$$

$$(2+2i)u = -a$$

$$\therefore a = -(2+2i)(2-i)$$

$$= -6-2i$$

$$\text{or } a = -(2+2i)(-2+i)$$

$$= 6+2i$$

Product of roots:

$$u(iu)(u+iu) = -b$$

$$b = u^3(1-i)$$

1

correct roots.

1

correct u'

MATHEMATICS Extension 2: Question...!

Suggested Solutions

Marks

Marker's Comments

$$\therefore b = (2-i)^3(1-i)$$

$$= -9-13i$$

$$\text{or } b = (-2+i)^3(1-i)$$

$$= 9+13i$$

$$\therefore x = 2b-13a$$

$$= 2(-9-13i)-13(-6-2i)$$

$$= 60$$

$$\text{or } x = 2(9+13i)-13(6+2i)$$

$$= -60$$

$$\therefore x = \pm 60$$

1

correct x value.

c) Assume

$$\frac{a+c}{\sqrt{a^2+c^2}} - \frac{b+c}{\sqrt{b^2+c^2}} \leq 0$$

1

correct assumption

$$\frac{a+c}{\sqrt{a^2+c^2}} \leq \frac{b+c}{\sqrt{b^2+c^2}}$$

$$(a+c)(\sqrt{b^2+c^2}) \leq (b+c)(\sqrt{a^2+c^2})$$

$$(a+c)^2(b^2+c^2) \leq (b+c)^2(a^2+c^2)$$

$$(a^2+2ac+c^2)(b^2+c^2) \leq$$

$$(b^2+2bc+c^2)(a^2+c^2)$$

$$\cancel{a^2b^2} + 2ab^2c + \cancel{c^2b^2} + \cancel{a^2c^2} + 2ac^3 + \cancel{c^4} \leq \cancel{a^2b^2} + 2a^2bc + \cancel{a^2c^2} + \cancel{b^2c^2} + 2bc^3 + \cancel{c^4}$$

$$\therefore 2ab^2c + 2ac^3 \leq 2a^2bc + 2bc^3$$

$$ab^2 + ac^2 \leq a^2b + bc^2$$

$$ab^2 + ac^2 - a^2b - bc^2 \leq 0$$

1

working

MATHEMATICS Extension 2: Question...10...

Suggested Solutions

Marks

Marker's Comments

$$ab(b-a) + c^2(a-b) \leq 0$$

$$\therefore (a-b)(c^2 - ab) \leq 0$$

$$\text{as } a > b \therefore (a-b) > 0$$

$$\text{and } c^2 > ab \therefore (c^2 - ab) > 0$$

$$\therefore (a-b)(c^2 - ab) > 0$$

$\therefore$ This is a contradiction

$$\text{hence, } \frac{a+c}{\sqrt{a^2+c^2}} - \frac{b+c}{\sqrt{b^2+c^2}} > 0$$

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correct explanation.

1

correct conclusion following correct explanation.

d) $A + B + C = \pi$ (interior angle sum of triangle)

$$A > 0, B > 0, C > 0$$

$$\frac{A+B+C}{3} \geq \sqrt[3]{ABC} \text{ (given)}$$

$$\therefore A+B+C \geq 3\sqrt[3]{ABC}$$

$$\frac{\frac{1}{A} + \frac{1}{B} + \frac{1}{C}}{3} \geq \sqrt[3]{\frac{1}{A} \times \frac{1}{B} \times \frac{1}{C}} \text{ (given)}$$

$$\therefore \frac{1}{A} + \frac{1}{B} + \frac{1}{C} \geq \frac{3}{\sqrt[3]{ABC}}$$

$$\therefore (A+B+C) \left(\frac{1}{A} + \frac{1}{B} + \frac{1}{C} \right) \geq 3\sqrt[3]{ABC} \times \frac{3}{\sqrt[3]{ABC}}$$

$$\pi \left(\frac{1}{A} + \frac{1}{B} + \frac{1}{C} \right) \geq 9 \text{ as } A+B+C=\pi$$

$$\therefore \frac{1}{A} + \frac{1}{B} + \frac{1}{C} \geq \frac{9}{\pi}$$

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